

NFL Pre-Snap Penalty Detector

Explainable computer vision prototype for first-pass review of broadcast football clips

PROJECT AT A GLANCE

Analyzes a football video frame by frame, detects people with pretrained YOLOv8, maintains lightweight player tracks, estimates the line of scrimmage and snap, and returns explainable suspected-penalty decisions. Manual offense, direction, LOS, and play-clock ROI inputs let a reviewer correct weak broadcast-camera assumptions.

IMPACT AND VALUE

- Speeds up first-pass clip review by turning motion and position histories into event flags.
- Covers false start, offside, delay of game, and too many men with evidence-rich reasons.
- Keeps decisions inspectable through track IDs, event frames, confidence, snap frame, and LOS.
- Portfolio value: integrates CV inference, tracking, OCR, temporal logic, CLI design, and tests.

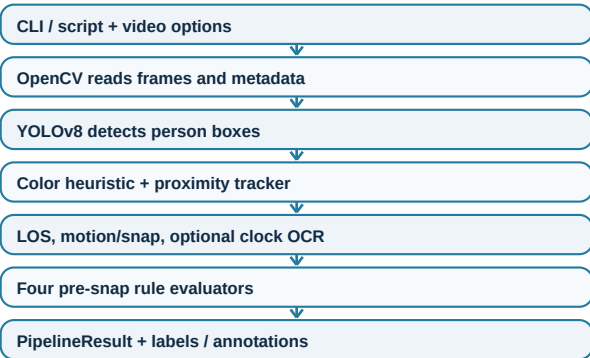
LANGUAGES AND STACK

Languages/assets: Python, YAML, TOML, Markdown, Jupyter, YOLO weights.

Libraries: OpenCV, Ultralytics YOLOv8, NumPy, pytesseract/Tesseract, pytest.

Design: dataclasses, modular pipeline, heuristic rules, optional human overrides.

CODE FLOW - TOP TO BOTTOM



FILE MAP - 38 FILES

Inventory: maintained/shipped tree, including stored weights; excludes .venv, caches, scratch, and test temp files.

.gitignore Excludes environments, secrets, weights, private video, outputs, and scratch files.	configs/default.yaml Minimal default model configuration scaffold.
data/README.md Explains where datasets and sample media belong.	data/annotations/README.md Marks the location for future annotation files.
demo/input/README.md Documents where authorized demo clips should be placed.	demo/output/README.md Documents the destination for generated demo artifacts.
LICENSE Repository license marker identifying MIT licensing.	models/README.md Explains where trained models or checkpoints belong.
nflpenaltydetector.py Legacy one-line Hello World starter retained in the tree.	notebooks/exploration.ipynb Empty notebook scaffold reserved for exploratory analysis.
pyproject.toml Declares project name, version, description, and Python 3.10+ requirement.	README.md Primary setup, CLI, method, data-rights, and limitation guide.
requirements.txt Lists OpenCV, Ultralytics, and pytesseract runtime packages.	scripts/evaluate.py Placeholder entry point for a future evaluation workflow.
scripts/run_demo.py Runs analysis on demo/input/sample.mp4 when the clip exists.	scripts/run_user_video.py Accepts a video path and prints core analysis fields.
scripts/run_video.py Placeholder entry point for a future generic video runner.	src/_init_.py Marks the root source directory as a Python package.
src/presnap_detector/_init_.py Initializes the detector package and exposes shared types.	src/presnap_detector/camera_motion.py Measures mean grayscale frame difference as a motion fallback.
src/presnap_detector/cli.py Parses video, offense, LOS, ROI, and Tesseract options; prints results.	src/presnap_detector/line_of_scrimmage.py Finds a blue-line contour centroid or falls back to mean player X.
src/presnap_detector/penalty_rules.py Infers offense and evaluates false start, offside, delay, and player count.	src/presnap_detector/pipeline.py Orchestrates detection, tracking, snap/LOS/OCR analysis, rules, and results.
src/presnap_detector/play_clock.py Crops and preprocesses the clock ROI, runs OCR, and parses digits.	src/presnap_detector/player_detector.py Loads YOLOv8n or custom weights and keeps person detections.
src/presnap_detector/snap_detector.py Builds normalized track-motion scores and selects the motion peak.	src/presnap_detector/team_classifier.py Labels blue-dominant crops home and red-dominant crops away.
src/presnap_detector/tracker.py Greeditly links nearby detections into short-gap tracks and team votes.	src/presnap_detector/types.py Defines Detection, PlayerTrack, and PenaltyDecision dataclasses.
src/presnap_detector/video_io.py Streams frames and reads frame count, dimensions, and FPS with OpenCV.	src/presnap_detector/visualization.py Draws detection boxes and labels on a frame.
tests/test_los_and_tracking.py Tests LOS estimation and proximity-track identity continuity.	tests/test_penalty_rules.py Tests all four supported suspected-penalty rules.
tests/test_play_clock.py Tests OCR text-to-clock-value parsing.	tests/test_snap_detector.py Tests peak-motion snap selection.
tests/test_video_io.py Creates a temporary clip and tests frame/property reads.	yolov8n.pt Stored pretrained YOLOv8 nano model weights used for person detection.